

VR-Forms. Preprint.

Part I. Foundations

Vitaly Reznik

2026

I.1. Place in the cycle

VR-Forms is the fourth work in the cycle developing the **VR** programme:

VR. A Formal System (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20212092) — a minimal axiomatisation of arithmetic on three primitives $\{\emptyset, \rightarrow, t\}$ and four axioms; equivalent to Peano arithmetic.

VR-Numbers (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20272743) — operational extensions over the natural numbers of VR: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ as operational procedures, without introducing pairs as objects.

VR-Sets (Reznik, 2026; Zenodo DOI 10.5281/zenodo.20303536) — an operational theory of sets: a set is a describable functionality, $\in$ is reference rather than containment. Russell's paradox dissolves ontologically. Countability of the universe as a consequence of the operational ontology.

VR-Forms is the present work. It introduces a second register — **formal terms**, descriptions without ontological force. These terms are formulas, and nothing more. They serve as a language in which VR can speak about everything that is not in its ontology: about classical uncountable objects, paradoxical classes, figures of myth and literature — about all descriptions that cannot be reduced to operational action.

Substantive connection with the preceding works: VR-Sets opened the boundary of operational ontology. VR-Forms describes how to speak about what lies beyond this boundary, without crossing it and without blurring it.

I.2. The initial position

The slogan of the whole cycle: *«only $\emptyset$ is, all else is doing»*. Applied to arithmetic, it gave VR; to numbers, VR-Numbers; to sets, VR-Sets. In all three works the slogan operated at the ontological level: the sole thing that is is $\emptyset$; everything else is operational action upon $\emptyset$.

VR-Forms adds a third type. Between **what is** ($\emptyset$) and **what acts** (operational functionalities), there is introduced **what is said**. Not as a new ontological entity — that would violate minimalism. Rather as a separate syntactic register, carrying no ontological commitments.

The full formulation of the position:

«Only $\emptyset$ is. Everything else either acts or is said. Action is operational and relates to $\emptyset$; speech is formal and refers to nothing by necessity.»

That is VR-Forms in a single sentence.

I.3. Two registers

We introduce two registers of reasoning. They differ not in subject matter and not in language, but in their relation to ontology.

The ontological register

This is the register of VR-Sets. The following principles hold:

- ontologically, only $\emptyset$ is;
- all other terms are operational functionalities, described by a finite procedure;
- the closure principle: every describable operational functionality is a set;
- contradictory descriptions do not specify functionalities and therefore do not specify sets.

In this register terms have reference: they refer to operational entities that are (in a derivative sense — as action upon $\emptyset$).

The formal register

This is the new register introduced by the present work. The following principles hold:

- terms are **formulas**;
- a formula specifies a **formal term**, and that is all it specifies;
- no formula is required to have reference;
- a formal term is a formal term and does not presuppose any other entity behind it.

In this register terms are speech. They refer to nothing by necessity. They are form — and only form.

Passage between registers

The same term may be considered in both registers. The description «the set of all functions $\mathbb{N} \rightarrow \mathbb{N}$ » in the ontological register requires verification: is it operational? Answer: no, the general set of all functions is not computable; hence in the ontological register it does not specify a set. The same description in the formal register is a formal term; no verification is required.

Passage between the registers is not an ontological operation. It is a passage between two ways of considering one and the same syntax. In one case we ask «is there a corresponding entity?»; in the other, we do not ask.

I.4. What a formal term is

A formal term is the syntactic record of a description. No more.

A formal term should not be thought of as a «potential set», or a «classical set to which VR has no access», or a «shadow object». All such readings are reifications of a formula,

attempts to ascribe reference to it. In VR-Forms this attempt is rejected as a matter of principle.

A formal term is form without content. Description without being. A record that describes — but the described does not exist, because in the formal register the question of existence is not posed.

This is not agnosticism. We do not say «perhaps it is, perhaps it is not». We say that the very question of existence is irrelevant in the formal register. To ask about the being of a formula is a category mistake. If we do wish to ask about being, we pass to the ontological register. There we obtain an answer: either «the corresponding operational functionality is» (and then the formal term corresponds to a VR-Sets set), or «not» (and then the formal term remains in the formal register and has no ontological correlate).

I.5. Examples of formal terms

Examples are divided into three groups, demonstrating the universality of the formal register.

Group 1. Mathematical formal terms

Classical $\wp(\mathbb{N})$ — the description «the set of all subsets of $\mathbb{N}$ ». In the ontological register of VR-Sets this is not a set in the full classical sense: VR contains a countable number of describable subsets of $\mathbb{N}$. In the formal register « $\wp(\mathbb{N})$ » is a formal term. The statement «the classical cardinality of $\wp(\mathbb{N})$ is $2^{\aleph_0}$ » belongs to the formal register and has no ontological force.

Classical $\mathbb{R}$ — the description «the set of all real numbers». In the ontological register $\mathbb{R}_{VR}$ is the countable set of computable reals. In the formal register «classical $\mathbb{R}$ » is a formal term distinct from $\mathbb{R}_{VR}$.

Russell's class — the description $\{x : x \notin x\}$. In the ontological register it is contradictory and does not specify a set (VR-Sets §II.3). In the formal register $R = \{x : x \notin x\}$ is a formal term. Russell's theorem now takes the form: R is a formal term having no ontological correlate in VR-Sets.

The Vitali set — the description «a set of representatives of equivalence classes on $\mathbb{R}$ under the relation $x - y \in \mathbb{Q}$ ». Requires choice over an uncountable family. Not operational. In the formal register a formal term.

Group 2. Paradoxical formal terms

The liar — the description «a statement equivalent to its own negation». In VR-Sets §VII.2 it was shown that in the ontological register it has no truth value. In the formal register the liar is a formal term; paradoxicality is a property of form, not an ontological failure.

The set of all sets — the description «an object whose elements are all sets». In the ontological register contradictory. In the formal register a formal term.

Group 3. Non-mathematical formal terms

The apparatus of the formal register is not confined to mathematics.

A dragon — the description «a fire-breathing lizard-like creature of large size with wings». This is a formula, a description. In the ontological register of VR-Sets — no operational functionality over $\emptyset$. In the formal register — a formal term.

God in monotheistic understanding — the description «the unique omnipotent eternal creator of the world». A formula. A formal term.

Unicorn. Phoenix. Centaur. Minotaur. Leviathan. All mythological beings are formal terms.

Soul. Idea in the Platonic sense. Universal. Essence. All classical philosophical terms are formal terms.

This last point has a fundamental significance. **The formal register of VR-Forms covers speech without exception.** Everything that is spoken of but is not operational is a formal term.

This is not reductionism. We do not say that dragons do not exist, or that God does not exist, or that universals do not exist. We say only: in the VR-ontology they are not; in the formal register one can speak of them as formal terms. The question of their existence outside the VR-ontology is not a mathematical question and not a question of the present work.

I.6. What VR-Forms does with speech

The introduction of the formal register changes the status of classical contested questions.

The question of the reality of the uncountable. In the VR-ontology the uncountable is not — the operational universe is countable (VR-Sets §VI.1). Classical mathematics freely uses the uncountable. VR-Forms gives a precise formulation: classical $\mathbb{R}$ is a formal term. Statements of classical mathematics about $\mathbb{R}$ are statements of the formal register. The dispute about the «reality» of the uncountable is a dispute about whether the corresponding formal term has an ontological correlate beyond VR. Mathematics does not answer this question; it shows only that in the VR-ontology there is none.

The question of paradoxes. Russell, the liar, Berry — all of them show: the corresponding formal terms have no ontological correlate in VR. This is the content of paradoxicality: the paradoxical description remains in the formal register. No damage to the VR ontology is done.

The question of the reality of non-mathematical entities. Dragons, gods, unicorns. In the VR-ontology they are not. In the formal register one can speak of them. Whether they exist in some other ontology is not a question of VR-Forms. The work only provides a formal language permitting talk of them without ontological commitment.

VR-Forms does not deny the existence of anything. It distinguishes what is operational in its own ontology from what is merely said and may (in another ontology) exist otherwise. About the first — ontological speech. About the second — formal speech.

I.7. Connection with the philosophical tradition

The position of VR-Forms has precedents in the history of philosophy.

Medieval nominalism (Ockham, Buridan). Universals are names, not entities. Only concrete individuals exist. VR-Forms inherits this position: formal terms are names; only the operational is real. The difference: for Ockham reality is ascribed to material individuals; in VR — only to $\emptyset$ and actions upon it. A more strict nominalism.

Mathematical constructivism (Brouwer, Bishop). Only that which is constructed exists. VR-Forms agrees with this in the ontological register. The difference: VR-Forms adds the formal register, in which one may speak of the non-constructive without claiming its existence. Classical constructivism refuses talk of the non-constructive; VR-Forms permits the talk but deprives it of ontological force.

Formalism (Hilbert). Mathematics is manipulation of formulas without reference. VR-Forms inherits formalism in the formal register. The difference: VR-Forms does not reduce all mathematics to the formal register. Part of mathematics has operational ontology — this is VR-Sets. Formalism deprives all mathematics of reference; VR-Forms — only its uncountable part.

Fictionalism (Field, Balaguer). Mathematical objects are useful fictions, not existing in the proper sense. VR-Forms agrees with this regarding uncountable and paradoxical objects. The difference: VR-Forms provides a formal apparatus for working with fictions, not only a philosophical position.

Late Wittgenstein. «Whereof one cannot speak, thereof one must be silent» — but be silent in a metaphysical sense, not a linguistic one. One may speak, while being aware that the speech is not a description of reality. VR-Forms continues this line: the formal register is speech aware of its referentlessness.

VR-Forms does not introduce anything unprecedented. It conducts a known philosophical position through the formal apparatus of VR. What is new in it is the combination of ontological minimality ($\emptyset$ as the sole primitive) with formal openness (any description is a formal term).

I.8. Structure of the work

The work consists of eight parts.

Part I. Foundations. Place in the cycle, initial position, two registers, formal term, examples (including non-mathematical), connection with the philosophical tradition.

Part II. The formal language. Precise definition of a formal term. Syntax of the formal register. Rules of formation of formal terms. Difference between a formal term and an operational set.

Part III. The principle of forms and consistency. The principle parallel to the closure principle of VR-Sets: every description specifies a formal term. Consistency of the extension.

Part IV. Transits. When work in the formal register yields results of the ontological register. The transit rule. The notion of witness. Safe and problematic transits.

Part V. Forms of mathematics. Application to mathematical formal terms: $\mathbb{R}$, $\wp(\mathbb{N})$, Vitali, paradoxical classes. What is translated into the VR-ontology, what remains in the formal register.

Part VI. Forms of speech. Application to non-mathematical formal terms: dragons, gods, philosophical categories. Universality of the formal register.

Part VII. Two-register logic. Logical properties of the extended system. Preservation of classical logic in both registers. Difference in rules of inference.

Part VIII. Open questions. Connection with proof theory. Preparation for VR-Audit. Possible applications of the formal register in other areas.

I.9. A note on tone

The work touches both the foundations of mathematics and speech in general. It is mathematical in apparatus and philosophical in scope. The tone is sustained in the manner of the previous works of the cycle: precision of definitions, sparsity of rhetoric, transparency of motivation.

The appearance of «dragons» and «gods» as examples of formal terms is not an artistic device or provocation. It is a direct consequence of the position: the formal register is universal. If it really covers speech without exception, it must be demonstrated on speech without exception.